Embedding format — asymmetric + raw query

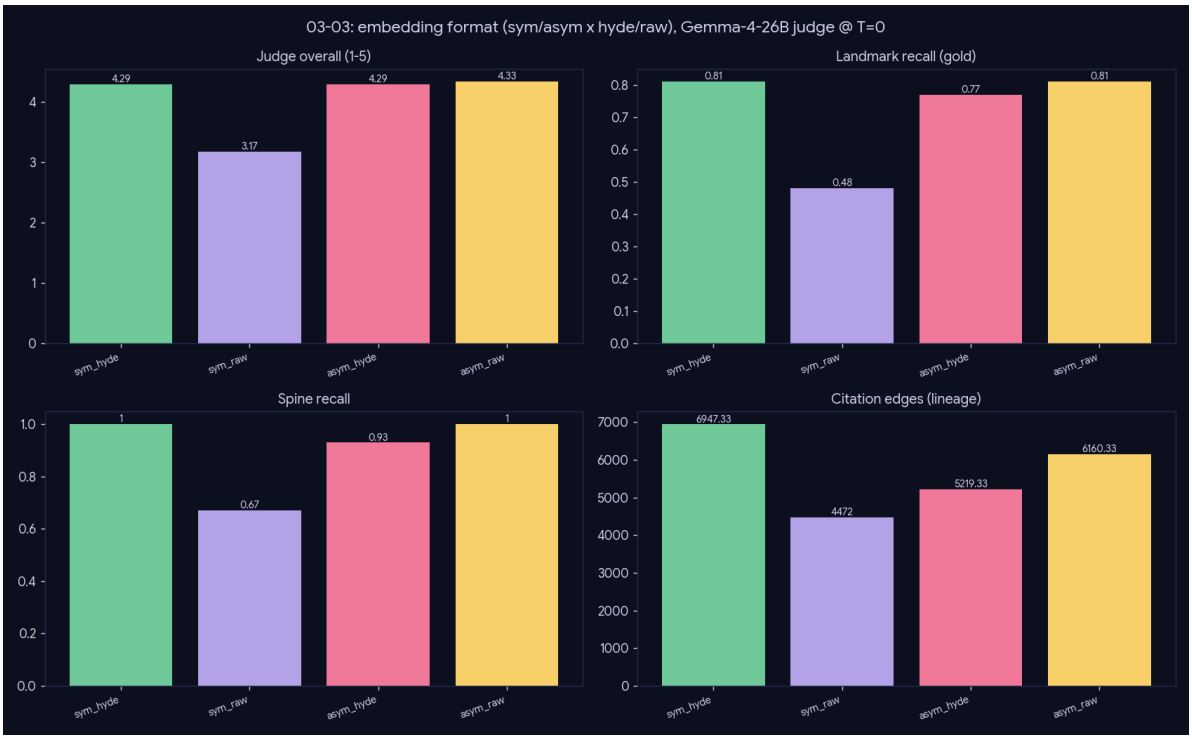
arxivkg · generation · embedding format

What we did

Four query-embedding conditions, full pipeline each (graph drill + PPR + compose), judged end-to-end over the 3 gold topics: `sym`/`asym` (symmetric vs nomic asymmetric `search_query`/`search_document`) × `hyde`/`raw` (a HyDE hypothetical-document expansion vs the raw question). The committed choice is `asym_raw`.

Results (3 gold topics)

format	n landmarks	n edges	grounded	off digest	landmark recall	spine recall	j grounding	j cov
sym_hyde	40.0	6947.33	22.67	0.67	0.81	1.0	4.33	
sym_raw	40.0	4472.0	13.67	0.0	0.48	0.67	3.0	
asym_hyde	40.0	5219.33	23.33	0.33	0.77	0.93	4.33	
asym_raw	40.0	6160.33	23.67	0.67	0.81	1.0	4.33	



Conclusion

The committed `asym_raw` ties for best — landmark recall 0.81, spine recall 1.0, top judge — at *no* extra generation. **`sym_raw` is the worst** (recall 0.48): a symmetric embedding on a bare question

under-retrieves the document space. **HyDE rescues the symmetric space** (`sym_hyde` recall 0.81, matching asym) but **adds nothing to the asymmetric space** (`asym_hyde` $\leq$ `asym_raw`) while costing a hypothetical-document generation per query. So the asymmetric query/document encoding already does what HyDE does — aligns the query into document space — making HyDE redundant. **Ship** `asym_raw`: best recall, no HyDE latency. **Caveat:** n= 3 gold; the judge axes are compressed, but recall/spine (objective) agree with the judge here.